

CompTIA A+ Core 1 Playbook

IT Labs

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2.4 Set up an A/V Workstation

The media department has recently hired a new video editor, and the existing workstation needs to be upgraded. Currently, the system is not equipped to meet the demands of video editing. It only has a single monitor connected to the on-board graphics adapter and limited storage capacity with no redundancy.

The workstation needs to support dual monitors and include a reliable storage solution that ensures data integrity, fault tolerance, and optimized performance for handling large video files. Configure the storage system to distribute data across multiple drives in a way that combines performance and redundancy, ensuring that the system can continue to function even if one drive fails.

When configuring the storage system, you will need to select all disks with the **model number WD800JD**.

Complete this lab as follows:

1. Select and install the video card.
 1. Above the computer, select the change view icon, then select **Motherboard** to switch to the motherboard view of the computer.
 2. Under Inventory, expand **GPUs/Expansion Cards**.
 3. Drag the **Video Adapter, DVI-I, HDMI, Crossfire, PCIe** to the PCIe slot on the computer.
 4. Select the **power supply**.
 5. From the Selected Item pane, drag the **Connector, Power Supply, 6-pin PCIe** power connector into the video card power slot.
 6. Select **GPUs/Expansion Cards** to return to the main Inventory list.
2. Install three hard drives (the minimum needed) for the RAID5 array.
 1. Under Inventory, expand **Storage**.
 2. Drag all three **hard drives** to a free 3.5" drive bay.
 3. Under Inventory, expand **Cables**.
 4. Select a **Cable, SATA**, and from the *Selected Item* pane:
 - Drag an unconnected **SATA connector** to a hard drive.
 - Drag the other unconnected **SATA connector** to an open SATA port on the motherboard.
 5. Repeat these steps to connect cables to the other two hard drives.
 6. Select the *partial Connections* icon by the computer, then select the **power supply**.
 7. From the Selected Item pane, drag an unconnected **SATA power connector** to a hard drive.
 8. Repeat the process for the remaining drives.
 9. Select **Cables** under Inventory to return to the main list.
3. Connect the original monitor to the new video card.
 1. Above the computer, select the change view icon, then select **Back**.
 2. On the computer, drag the **DVI connector** from the built-in video connector to the new video card in the computer.
4. Connect a second monitor.
 1. Under Inventory, expand **Peripherals**.
 2. Drag the **Monitor** to the left of the desktop.
 3. Above the left monitor, select the change view icon, then select **Back**.
 4. Under Inventory, expand **Cables**.
 5. Select the **HDMI to HDMI Cable**.
 6. From the Selected Item pane:
 - Drag an **HDMI Connector** to the new video card in the computer.
 - Drag the unconnected **HDMI Connector** to the HDMI port on the left monitor.

7. Under Inventory, select the **AC Power Cable**.
8. From the Selected Item pane:
 - Connect the **Female AC Power Connector** to the left monitor.
 - Connect the **Male AC Power Connector** to a wall outlet.
9. Above the left monitor, select the change view icon, then select **Front**.
10. Select the **Power button** to power on the left monitor.
5. Power on the computer and enable RAID in the BIOS settings.
 1. Above the computer, select the Change View icon, then select **Front**.
 2. Select the **power** button to turn on the computer.
 3. As the computer boots, press **F2** (or **Delete**) to enter the BIOS setup program.
 4. From the left menu, expand **System Configuration**.
 5. Select **SATA Operation**.
 6. From the right pane, select **RAID On**.
 7. Select **Apply**.
6. Exit BIOS and configure the RAID array.
 1. Select **Exit** to restart the system.
 2. After the system boots and the BIOS loads, the RAID controller will load.
 3. When the RAID message screen displays, press **Ctrl + I** (i).
 4. With *Create RAID Volume* highlighted, press **Enter**.
 5. On the *Name* field, press **Enter**.
 6. On the RAID Level field, use the **up and down arrow keys** to select **RAID5 (Parity)**.
 7. Press **Enter**.
 8. With *Select Disks* highlighted, press **Enter**.
 9. Use the **up and down arrow keys** to traverse the list and the **spacebar** to select a drive.
 10. Select all three drives with the *model number* *WD800JD* and press **Enter**.
 11. Press the **Tab** key.
 12. Verify that the *Strip size* is **128 KB**. If needed, use the up and down arrows to set the value.
 13. Press **Enter**.
 14. Press **Enter** to accept the default capacity.
 15. With *Create Volume* highlighted, press **Enter**.
 16. Press **Y** to continue creating the volume.
 17. Press the escape key (**Esc**).
 18. Press **Y** to exit.